# Natural language processing

# Text to persons

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# Entity recognition

Entity recognition: Given a string identify the parts of the string that reference an entity. An entity could be a person, an organization, a location.

Example:

| Text: | Finn | is | on | DTU | in | Lyngby |
|---|---|---|---|---|---|---|
| Category: | PER | | | ORG | | LOC |

Example with multiword string

| Finn | is | on | the | Technical | University | of | Denmark | in | Lyngby |
|---|---|---|---|---|---|---|---|---|---|
| PER | | | | ORG BEGIN | — | — | ORG END | | LOC |

# Entity linking

Entity recognition is the first step towards entity linking.

Entity linking is there you link a string that to a data resource item, e.g., the knowledge graph Wikidata.

Example:

| | | | | | | |
|---|---|---|---|---|---|---|
| Text: | Finn | is | on | DTU | in | Lyngby |
| Category: | PER | | | ORG | | LOC |
| Linking Wikidata QID: | Q20980928 | | | Q1269766 | | Q2239 |

# Prompt enginering

Normal

"Chain of thought"

Zero, one shot, few shot: Using examples to demonstrate how the

Metaprompt: A prompt to make prompts

# DSPy

Python library which encapsulates working with prompts a converting response.

Meta prompting

Optimization of prompt

Optimization of which examples should be used in few shot prompting

Classes for specification of input and output.

ReAct: Reasoning-action where LLM processing are interleaved with API calls (or other actions.

# CampusAI

CampusAI provides a chat interface and chat API and embedding API.

Available from https://campusai.compute.dtu.dk

Set up in the autumn 2025 and being developed, with models sometimes change. Stable are probably Gemma3 (generation) and nomic-embed-text (embedding).

Free for users with login on DTU. You need to be on the DTU network.

OpenAI-compatible: you can use the `openai` python package and set the `base_url` to the appropriate URL for CampusAI.

API key available in the interface. Add it to the `~/.env` file. Never add it to the Python code that you checkinn in a public repository.

If CampusAI is overloaded try a local Ollama.